University of Warwick
MSFT

IB9AU0 Generative AI and AI Applications
2025-2026

Individual Assignment 2 Question

1500-word essay:

Objective: build a robust, modern Retrieval-Augmented Generation (RAG) system in a domain of your choice that benefits from RAG (i.e., domain with specialized or up-to-date knowledge). Demonstrate improvements over a baseline LLM.

Requirements (must include all):

1. **Domain & Data Sourcing**
    - Choose a specific domain or sub-domain (e.g. regulatory texts, company filings, product docs, research papers, niche FinTech topics, etc.).
    - Source a dataset or knowledge base (public or legitimately shared) that contains information unlikely to be in standard LLM training data (e.g. recent regulation, very niche disclosures, local-market documents).
    - Provide a short justification: why this domain needs RAG rather than just prompting.
2. **Research & Implement Recent/Advanced RAG Improvements (at least one):**
    - E.g. hybrid retrieval (keyword + vector), re-ranking model for retrieval, dynamic chunking, semantic / hierarchical chunking, LLM-based query rewriting or HyDE, self-refinement loops, LLM-augmented search, etc.
3. **Complete System Implementation:**
    - Preprocessing: chunking, embedding, vector store.
    - Retrieval & reranking.
    - Generation with retrieved context (prompt engineering, grounding).
    - Baseline pipeline (e.g. simple retrieval + vanilla prompt) + your enhanced pipeline.
4. **Evaluation:**
    - Design a test set of queries relevant to your domain (diverse: simple facts, deep context, ambiguous questions, edge cases).
    - Evaluate both retrieval (precision/recall) and generation (correctness, grounding, completeness).
5. **Analysis & Reflection:**
    - Compare baseline vs your enhanced system.
    - Identify failure modes.
    - Propose realistic improvements and limitations (e.g. latency, hallucinations, data gaps, update strategy, maintenance).

Deliverables:

- Notebook(s) / code for full pipeline.
- A report describing design decisions, results, evaluation, and reflection.
- A short “demo log” — sample inputs + system outputs + commentary.

Marking Criteria
Completeness of the solution, evidence of consideration of best practices in the given domain, efficiency of the code, interpretation of the results,

Submission Instructions
The assignment solutions should be submitted as a zip file containing both the written narrative for each part as well as the code in the form of python code (either .py files or Jupyter notebooks). The students should combine all files with a zip file with the following naming format:

student_number.zip

No other files are going to be considered. It is your responsibility to comply with the requirements of the submission, otherwise this will have repercussions for marking.

Maximum Length:	**1500 words (this is a strict limit, not a guideline)**
Assessment Weighting:	**50%**
Submission Deadline:	**16/04/2026 12:00 PM (UK time)**
Marks Released by:	**18/05/2026** (we will aim to release marks by this date, but in the event of an unavoidable delay a message will be posted on the module page)

Use of Artificial Intelligence

The University recognises an increasing number of technologies such as Artificial Intelligence and that they may be applicable in your completing this assessment. The assessment brief sets out specific requirements or restrictions, and your student handbook has further guidance and advice.

You are reminded that the inappropriate use of such a technology may constitute a breach of University policy, such as the Proofreading Policy or Regulation 11 (Academic Integrity). If you breach these policies, it may have significant consequences for your studies. Please make sure you read and understand the assessment brief and how AI may or may not be used.

You should ensure you include an assignment cover sheet which requires you to explain the use of any AI. Failure to disclose at the point of submission may be prejudicial in any later investigations should they arise.

For this assessment, AI is*:* **REQUIRED**

You MUST set out clearly in your submission how you used the tool/service and You MUST set out clearly what output from the tool/service has been included, and where you have altered, adopted or built on that output.

Submitting your work

Before you submit your assessment, you should ensure you are familiar with the guidance and rules in the "Your Assessments" section of your Student Handbook, paying particular attention to:

- Inserting a completed assignment coversheet as the first page of your submission
- Academic Integrity (including Plagiarism)
- Referencing
- Word Count Policy and Formatting
- Confidentiality of your work
- Extensions and late submission of work
- Guidelines for Online Submission

Mitigating Circumstances

Mitigating circumstances MUST be submitted **within 20 working days following the submission deadline**. Mitigating circumstances not submitted by the relevant deadline are not required to be considered by the School/Department and may have to be considered by an Academic Appeals Committee as part of an academic appeal – for further information, please see: https://warwick.ac.uk/services/gov/calendar/section2/regulations/reg42academicappeals

Please see your Student Handbook for more guidance on mitigating circumstances